

频率法串联超前校正

例 1 以知某单位反馈系统的开环传递函数为：

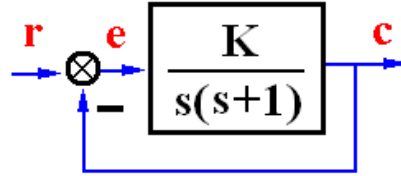
$$G(s) = \frac{K}{s(s+1)}$$

(I) 要求： $r(t) = t$ 时， $e_{ss}^* \leq 0.1$

$$\omega_c^* \geq 5$$

$$\gamma^* \geq 60^\circ$$

$$h^* \geq 10 \text{ dB}$$



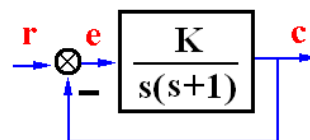
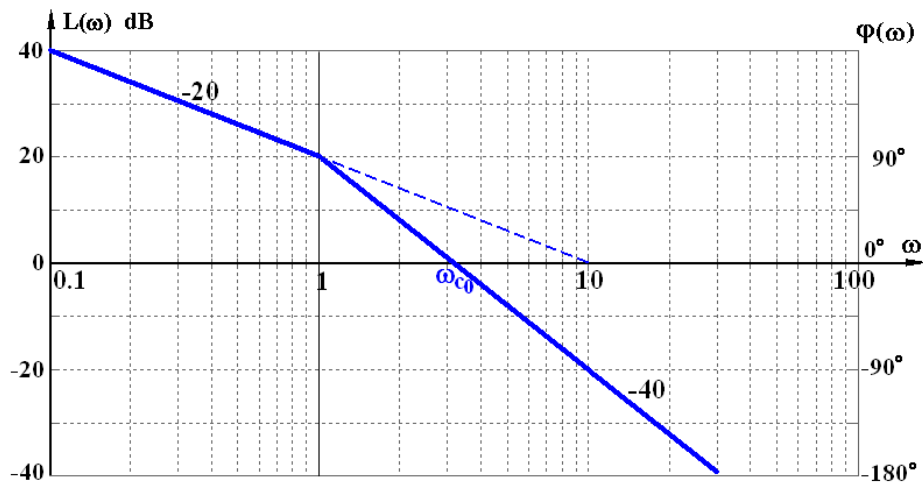
试确定串联校正装置的传递函数 $G_c(s)$ 。

解： $G(s) = \frac{K}{s(s+1)} \quad \begin{cases} K = K \\ v = 1 \end{cases}$

① 根据稳态误差要求确定 K ；

$$e_{ss}^* = \frac{1}{K} \leq 0.1, \quad \text{取 } K = 10。$$

② 作 $L_0(\omega)$ 特性曲线如图所示：



$$\omega_{c0} = \sqrt{10} = 3.16 \quad (< \omega_c^* = 5)$$

$$\gamma_0 = 180^\circ + \varphi(\omega_{c0})$$

$$= 180^\circ - 90^\circ - \arctan 3.16 = 17.6^\circ \quad (< \gamma^* = 60^\circ)$$

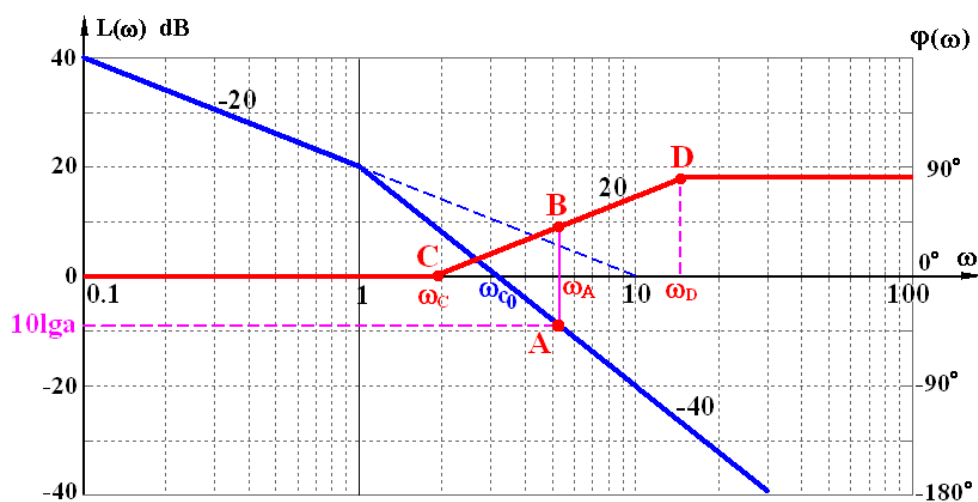
③ 确定采用串联超前校正

$$\varphi_m = \gamma^* - \gamma_0 + (5^\circ \sim 10^\circ)$$

$$= 60^\circ - 17.6^\circ + 5^\circ = 47.4^\circ$$

$$\begin{cases} a = \frac{1 + \sin 47.4^\circ}{1 - \sin 47.4^\circ} = 6.58 \approx 7 \\ 10 \lg a = 8.5 \text{ dB} \end{cases}$$

④ 确定 $G_c(s)$ —— $10 \lg a = 8.5 \text{ dB} \rightarrow A, B, C, D$



$$A: 8.5 = 40 \cdot \lg \frac{\omega_A}{\omega_{c0}}, \quad 10^{\frac{8.5}{40}} = \frac{\omega_A}{\omega_{c0}}, \quad \omega_A = \omega_{c0} \cdot 10^{\frac{8.5}{40}} = 5.16 > 5 = \omega_c^*$$

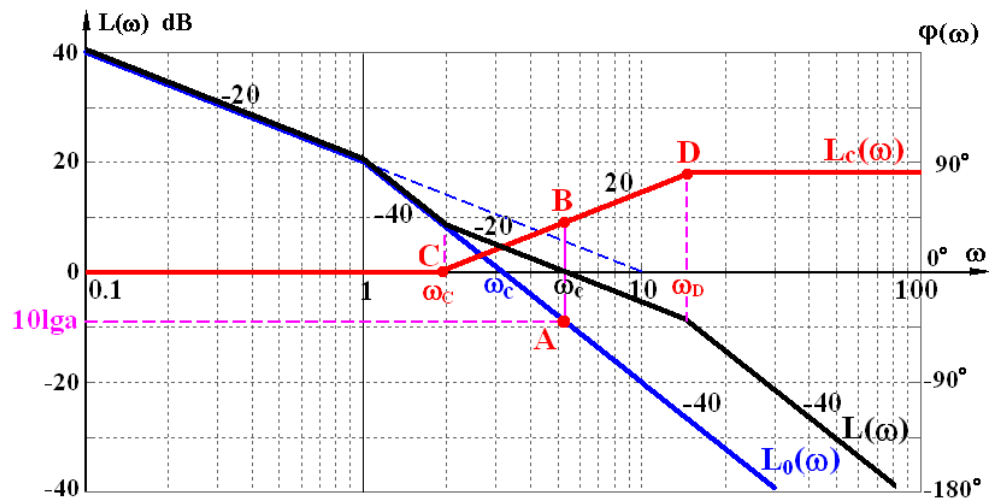
$$C: \frac{\omega_A}{\omega_{c0}} = \frac{\omega_{c0}}{\omega_C}, \quad \omega_C = \frac{\omega_{c0}^2}{\omega_A} = \frac{3.16^2}{5.16} = 1.94$$

$$D: \frac{\omega_D}{\omega_A} = \frac{\omega_A}{\omega_C}, \quad \omega_D = \frac{\omega_A^2}{\omega_C} = \frac{5.16^2}{1.94} = 13.73$$

$$G_{c1}(s) = \frac{\frac{s}{\omega_C} + 1}{\frac{s}{\omega_D} + 1} = \frac{\frac{s}{1.94} + 1}{\frac{s}{13.74} + 1}$$

$$\begin{cases} T = 1/13.74 = 0.0728 \\ aT = 1/1.94 = 0.56 \\ a = 13.74/1.94 = 7.09 \end{cases}$$

$$G(s) = G_{c1}(s) \cdot G_0(s) = \frac{10 \cdot \left(\frac{s}{1.94} + 1\right)}{s(s+1)\left(\frac{s}{13.74} + 1\right)}$$



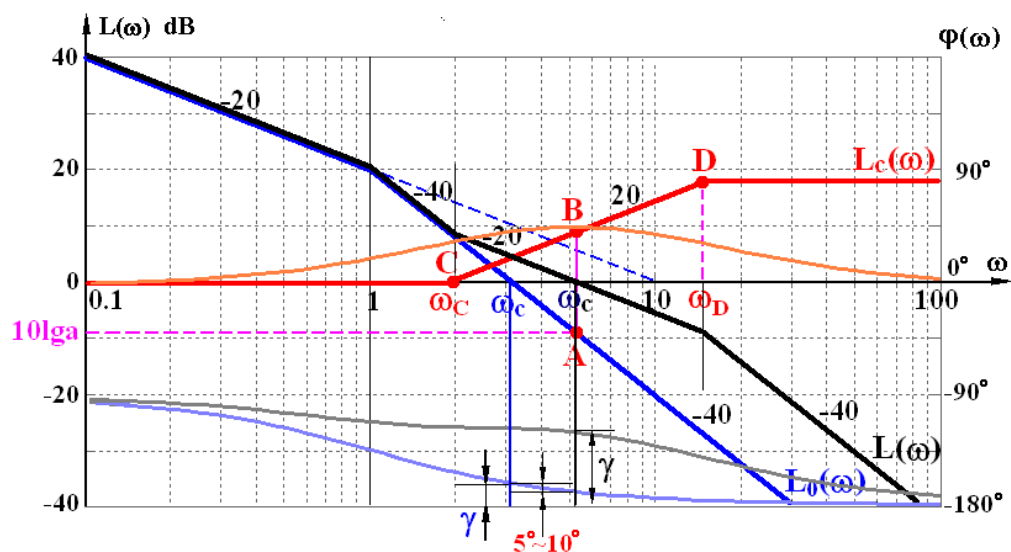
$$G(s) = G_{c1}(s) \cdot G_0(s) = \frac{10 \cdot \left(\frac{s}{1.94} + 1\right)}{s(s+1)\left(\frac{s}{13.74} + 1\right)}$$

验算 1: $e_{ss} = \frac{1}{K} = \frac{1}{10} = 0.1$

$$\omega_{c1} = \omega_A = 5.16 \quad (> 5)$$

$$\gamma = 180^\circ + \varphi(\omega_{c1})$$

$$\begin{aligned} &= 180^\circ + \arctan \frac{5.16}{1.94} - 90^\circ - \arctan 5.16 - \arctan \frac{5.16}{13.73} \\ &= 180^\circ + 69.4^\circ - 90^\circ - 79^\circ - 20.6^\circ = 58.8^\circ \quad (< 60^\circ) \end{aligned}$$



修改：取 $G_{c2}(s) = \frac{\frac{s}{1.94} + 1}{\frac{s}{15} + 1}$ $\begin{cases} T = 1/15 = 0.0667 \\ a = 15/1.94 = 7.732 \end{cases}$

$$G(s) = G_{c2}(s) \cdot G_0(s) = \frac{10(\frac{s}{1.94} + 1)}{s(s+1)(\frac{s}{15} + 1)}$$

$$\begin{aligned} \gamma_2 &= \gamma_1 + \arctan \frac{5.16}{13.74} - \arctan \frac{5.16}{15} \\ &= 58.8^\circ + 20.6^\circ - 19^\circ = 60.4^\circ \quad (> 60^\circ) \end{aligned}$$

$$h^* = \infty \quad (> 10 \text{ dB})$$

验算： $\begin{cases} e_{ss} = 0.1 \\ \omega_c = 5.16 > 5.0 \\ \gamma = 60.4^\circ > 60^\circ \\ h = \infty > 10 \text{ dB} \end{cases}$ 校正完成。



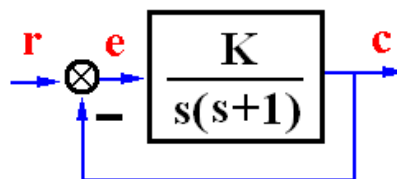
(II) 按 $\omega_c^* \geq 7$ 进行设计

要求: $r(t) = t$ 时, $e_{ss}^* \leq 0.1$

$$\omega_c^* \geq 7$$

$$\gamma^* \geq 60^\circ$$

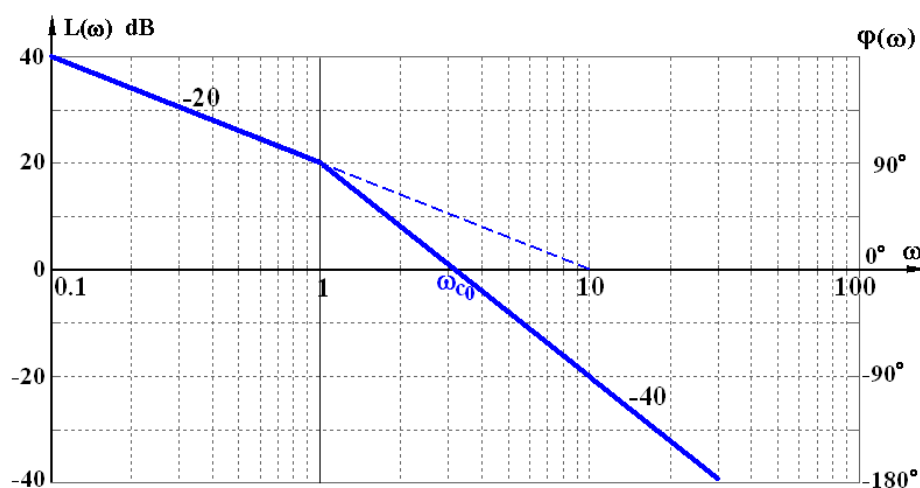
$$h^* \geq 10 \text{ dB}$$



① 据稳态误差要求确定 K ;

$$e_{ss}^* = \frac{1}{K} \leq 0.1, \text{ 取 } K = 10$$

② 绘制 $L_0(\omega)$ 曲线如图所示:



$$\omega_{c0} = \sqrt{10} = 3.16 \quad (< \omega_c^* = 5)$$

$$\gamma_0 = 180^\circ + \varphi(\omega_{c0})$$

$$= 180^\circ - 90^\circ - \arctan 3.16 = 17.6^\circ \quad (< \gamma^* = 60^\circ)$$

③ 确定采用串联超前校正

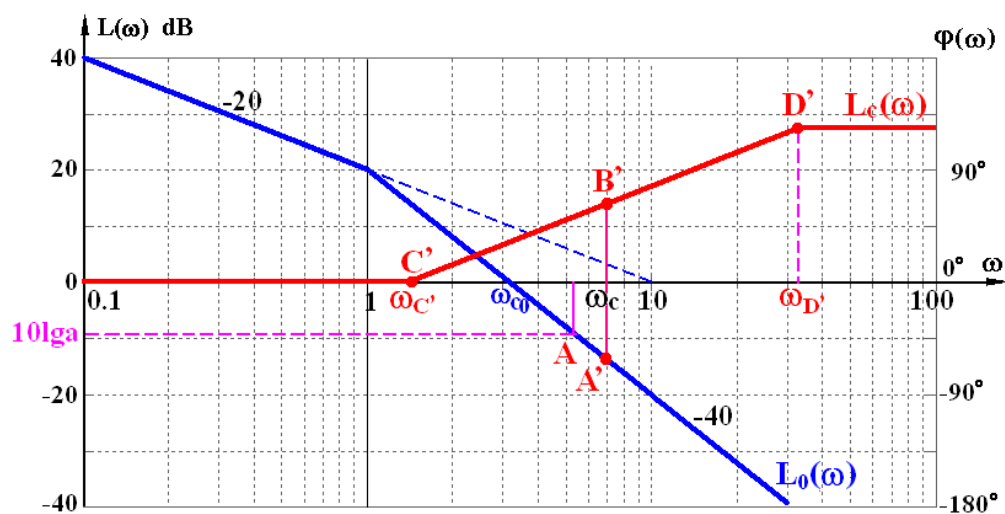
$$\varphi_m = \gamma^* - \gamma_0 + (5^\circ \sim 10^\circ)$$

$$= 60^\circ - 17.6^\circ + 5^\circ = 47.4^\circ$$

$$\begin{cases} a = \frac{1 + \sin 47.4^\circ}{1 - \sin 47.4^\circ} = 6.58 \approx 7 \\ 10 \lg a = 8.5 \text{ dB} \end{cases}$$

④ 确定 $G_c(s)$ —— $10 \lg a = 8.5 \text{ dB} \rightarrow A, B, C, D$

在 $\omega_{A'} = 7$ 处划竖线: $\rightarrow A', B', C', D'$



$$A': \omega_{A'} = \omega_c = 7 = \omega_c^*$$

$$C': \frac{\omega_c}{\omega_{c0}} = \frac{\omega_{c0}}{\omega_{C'}}, \quad \omega_{C'} = \frac{\omega_{c0}^2}{\omega_c} = \frac{3.16^2}{7} = 1.43$$

$$D': \frac{\omega_{D'}}{\omega_c} = \frac{\omega_c}{\omega_{C'}}, \quad \omega_{D'} = \frac{\omega_c^2}{\omega_{C'}} = \frac{7^2}{1.43} = 34.3$$

$$G_c(s) = \frac{\frac{s}{\omega_{C'}} + 1}{\frac{s}{\omega_{D'}} + 1} = \frac{\frac{s}{1.43} + 1}{\frac{s}{34.4} + 1}$$

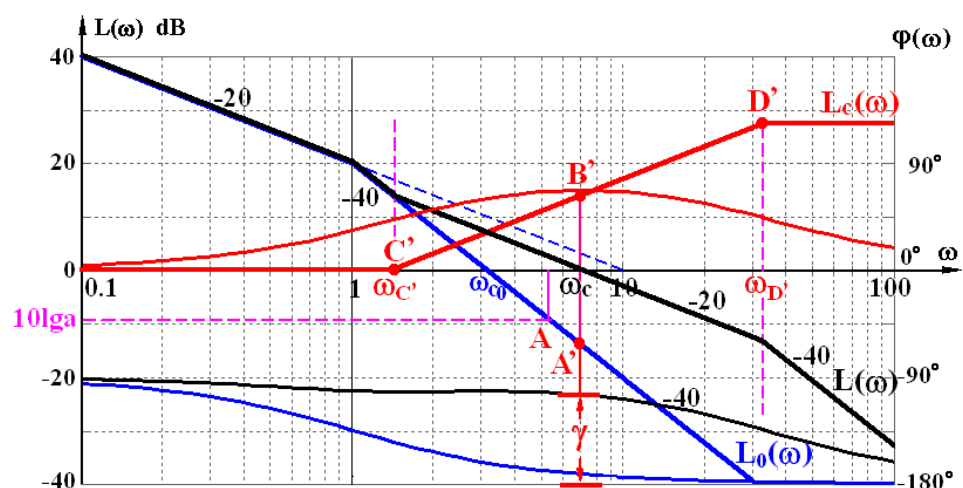
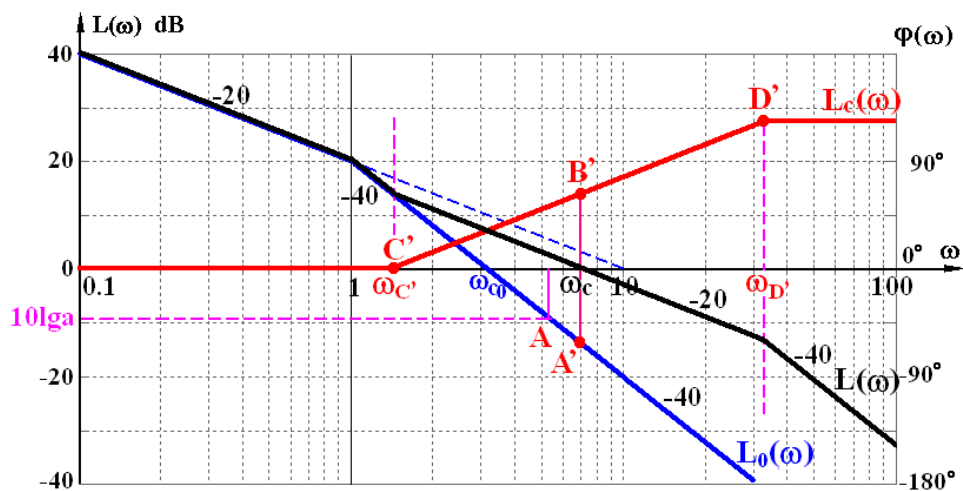
⑤ 验算: 校正后开环传递函数

$$G(s) = G_c(s) \cdot G_0(s) = \frac{10 \cdot (\frac{s}{1.43} + 1)}{s(s+1)(\frac{s}{34.3} + 1)}$$

$$\begin{aligned} \gamma &= 180^\circ + \arctan \frac{7}{1.43} - 90^\circ - \arctan 7 - \arctan \frac{7}{34.3} \\ &= 180^\circ + 78.5^\circ - 90^\circ - 81.9^\circ - 11.5^\circ = 75.1^\circ \quad (> 60^\circ) \end{aligned}$$

$$h^* = \infty \quad (> 10 \text{ dB})$$

$$\text{验算结果: } \begin{cases} e_{ss} = 0.1 \\ \omega_c = 7 = \omega_c^* \\ \gamma = 75.1^\circ > 60^\circ \\ h = \infty > 10 \text{ dB} \end{cases} \quad \text{校正完成。}$$



采用超前校正的效果：

中频段： ω_c ， γ 两项指标得以改善——动态指标 t_s ， $\sigma\%$ 变好了。

高频段： $G_c(s)$ 幅值增加， $L(\omega)$ 高频段抬高——抗高频噪声能力降低。